# Formal Proof of the Branch-Safety Lemma

for the eml$^\star$ Minimal Anti-Holomorphic Extension

Extension of Odrzywołek (arXiv:2603.21852v2) — Monnerot (2026)

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**Abstract**

We provide a formal mathematical proof of the branch-safety lemma underlying the eml$^\star$ operator. The lemma establishes the precise domain on which the principal-branch evaluation of eml$^\star$ recovers the complex conjugate without $2\pi i$ discontinuities. We prove necessity and sufficiency of the condition $\mathrm{Im}(z) \in [-\pi, \pi)$ and discuss the extension to general expression trees used in the density theorem (Theorem 4.3 of the companion paper).

## 1 Statement of the Branch-Safety Lemma

Recall the two operators introduced in Monnerot (2026):

$$\mathrm{eml}(x, y) \;:=\; e^x - \mathrm{Log}(y), \qquad \mathrm{eml}^\star(x, y) \;:=\; e^x - \mathrm{Log}(\overline{y}),$$

where Log denotes the *principal branch* of the complex logarithm, defined on $\mathbb{C} \setminus (-\infty, 0]$ with $\mathrm{Arg} \in (-\pi, \pi]$.

**Lemma 1.1** (Branch Safety for Conjugate Recovery). *Let $z \in \mathbb{C}$ satisfy $\mathrm{Im}(z) \in [-\pi, \pi)$. Then*

$$\overline{z} \;=\; 1 - \mathrm{eml}^\star\big(0,\, \mathrm{eml}(z, 1)\big),$$

*where both operators are evaluated using the principal branch of* Log.

*Moreover, the condition is* sharp*: the equality fails (by a jump of exactly $2\pi i$) when* $\mathrm{Im}(z) = \pi$.

## 2 Formal Proof

We prove the identity by explicit computation, tracking the imaginary part at each step.

**Step 1: Evaluation of the inner** eml**.** By definition,

$$\mathrm{eml}(z, 1) \;=\; e^z - \mathrm{Log}(1).$$

Since $\mathrm{Log}(1) = 0$ (the principal value; 1 lies safely away from the branch cut $(-\infty, 0]$), we obtain

$$\mathrm{eml}(z, 1) \;=\; e^z.$$

**Step 2: Conjugation of the exponential.** Set $w := \mathrm{eml}(z, 1) = e^z$. Then $\overline{w} = \overline{e^z}$. The exponential function commutes with complex conjugation for all $z \in \mathbb{C}$:

$$\overline{e^z} \;=\; e^{\overline{z}}.$$

This follows either from the power-series definition $e^z = \sum_{n=0}^{\infty} z^n/n!$ (the coefficients $1/n!$ are real, so conjugation passes inside the sum) or from the polar identity $e^{x+iy} = e^x(\cos y + i \sin y)$ together with $\cos(-y) = \cos y$ and $\sin(-y) = -\sin y$.

**Step 3: Evaluation of $\mathrm{eml}^\star$.** Now apply $\mathrm{eml}^\star(0, w)$:

$$\mathrm{eml}^\star(0, w) \;=\; e^0 - \mathrm{Log}(\overline{w}) \;=\; 1 - \mathrm{Log}\big(e^{\overline{z}}\big).$$

**Step 4: Inversion property of Log on the closed strip.** Set $u := \overline{z}$. Then $\mathrm{Im}(u) = -\,\mathrm{Im}(z)$. The hypothesis $\mathrm{Im}(z) \in [-\pi, \pi)$ immediately implies

$$\mathrm{Im}(u) \;=\; -\,\mathrm{Im}(z) \;\in\; (-\pi, \pi].$$

It is a standard result in complex analysis that the principal logarithm $\mathrm{Log} : \mathbb{C} \setminus (-\infty, 0] \to \{w : \mathrm{Im}(w) \in (-\pi, \pi]\}$ is a two-sided inverse of the restriction of exp to the strip $\{v : \mathrm{Im}(v) \in (-\pi, \pi]\}$: for every $v$ with $\mathrm{Im}(v) \in (-\pi, \pi]$,

$$\mathrm{Log}(e^v) \;=\; v.$$

Applying this with $v = u = \overline{z}$ (whose imaginary part lies in the *closed* interval $(-\pi, \pi]$) yields

$$\mathrm{Log}\big(e^{\overline{z}}\big) \;=\; \overline{z}.$$

**Step 5: Conclusion.** Substituting back:

$$\mathrm{eml}^\star\big(0, \mathrm{eml}(z, 1)\big) \;=\; 1 - \overline{z}.$$

Therefore

$$1 - \mathrm{eml}^\star\big(0, \mathrm{eml}(z, 1)\big) \;=\; \overline{z},$$

which is the claimed identity. □

## 3 Sharpness of the Domain

We prove that $[-\pi, \pi)$ is the largest connected interval of imaginary parts for which the formula holds.

**Theorem 3.1** (Exact Safe Domain). *The identity* $\overline{z} = 1 - \mathrm{eml}^\star(0, \mathrm{eml}(z, 1))$ *holds if and only if* $\mathrm{Im}(z) \in [-\pi, \pi)$.

*Proof. Sufficiency* is Lemma 1.1 (Section 2).

*Failure at* $\mathrm{Im}(z) = \pi$. Let $z = x + i\pi$ with $x \in \mathbb{R}$. Then $\overline{z} = x - i\pi$ and

$$e^{\overline{z}} \;=\; e^{x - i\pi} \;=\; -e^x \;<\; 0.$$

The principal logarithm on the negative real axis returns $\mathrm{Arg} = +\pi$:

$$\mathrm{Log}(-e^x) \;=\; x + i\pi.$$

But $\overline{z} = x - i\pi$, so

$$\mathrm{Log}\big(e^{\overline{z}}\big) \;=\; \overline{z} + 2\pi i \;\neq\; \overline{z}.$$

The formula therefore yields a value off by $2\pi i$ and fails.

*Failure for* $\mathrm{Im}(z) > \pi$. Let $\mathrm{Im}(z) = \pi + \varepsilon$ with $\varepsilon > 0$. Then $\mathrm{Im}(\overline{z}) = -\pi - \varepsilon < -\pi$, so $e^{\overline{z}}$ has argument $-\pi - \varepsilon \notin (-\pi, \pi]$. The principal Log wraps this to argument $\pi - \varepsilon$, introducing a jump of $2\pi i$ in the same way.

*Safety at* $\mathrm{Im}(z) = -\pi$. At the lower boundary, $\mathrm{Im}(\overline{z}) = \pi \in (-\pi, \pi]$, so $\mathrm{Log}(e^{\overline{z}}) = \overline{z}$ by Step 4, and the formula holds. This is confirmed by direct symbolic computation (SymPy): `simplify(1 - (1 - log(-exp(x)))) == x + I*pi` returns `True`.

Combining: the formula holds if and only if $\mathrm{Im}(z) \in [-\pi, \pi)$. □

**Remark 3.2.** *The boundary* $\mathrm{Im}(z) = -\pi$ *is mathematically safe but numerically unstable: floating-point evaluation of* $e^{-i\pi}$ *produces a tiny positive imaginary part, which can push* $\overline{e^{z}}$ *just below the branch cut, causing a spurious* $2\pi i$ *jump. The reliable numerical domain is therefore the open interval* $(-\pi, \pi)$*. See* `verify_theorem4.py` *for a detailed discussion.*

## 4 Extension to General Expression Trees

The grammar $S \to 1 \mid \mathrm{eml}(S,S) \mid \mathrm{eml}^\star(S,S)$ generates, when leaves include the variable $z$, a family of functions closed under composition. Each occurrence of $\mathrm{eml}^\star$ introduces a local conjugate recovery whose safety depends only on whether the imaginary part of its local argument lies in $[-\pi, \pi)$.

**Assumption 4.1** (Branch safety of arithmetic trees)**.** For the specific addition tree (RPN-length 19) and multiplication tree (RPN-length 17) of Odrzywołek [1], every intermediate argument of every eml-node has imaginary part in $(-\pi, \pi]$ whenever the global input satisfies $|\operatorname{Im}(z)| < 1$.

**Remark 4.2.** *Assumption 4.1 is verified numerically at 60 decimal places: zero violations were found on a grid and 1000 random samples inside $|\operatorname{Im}(z)| < 1$. A fully analytic interval-arithmetic proof — propagating Im-bounds forward through each node — is in principle obtainable but has not been written out in full; it remains an open verification task.*

By structural induction, if Assumption 4.1 holds for a given compact $K \subset \{z : |\operatorname{Im}(z)| < \pi\}$, then every composite tree built from eml, $\mathrm{eml}^\star$, and the constant 1 evaluates without branch-cut crossings on $K$.

## 5 Consequences for the Density Theorem

**Theorem 5.1** (Stone–Weierstrass Density, Th. 4.3 of companion paper)**.** *Under Assumption 4.1, for every compact $K \subset \{z : \operatorname{Im}(z) \in (-\pi, \pi)\}$, the algebra generated by $\{\mathrm{eml}, \mathrm{eml}^\star, 1\}$ is dense in $C(K, \mathbb{C})$.*

*Proof.* On such $K$, Lemma 1.1 is exact and continuous, so $\bar{z}$ belongs to the algebra pointwise on $K$. Combined with Assumption 4.1, the algebra contains $z$, $\bar{z}$, all rational constants, and is closed under addition and multiplication; hence it contains $\mathbb{Q}[z, \bar{z}]$.

The sub-algebra $A = \mathbb{Q}[z, \bar{z}] \subset C(K, \mathbb{C})$ satisfies the hypotheses of the complex Stone–Weierstrass theorem [2]:

(i) *Separates points*: for $z_1 \neq z_2$, $z \in A$ gives $z(z_1) \neq z(z_2)$.

(ii) *Does not vanish*: $1 \in A$.

(iii) *Closed under conjugation*: $z \in A$ and $\bar{z} \in A$ imply $\bar{f} \in A$ for any $f \in A$.

Hence $A$ is dense in $C(K, \mathbb{C})$. □

**Remark 5.2.** *Theorem 5.1 does not contradict Theorem 2.1 of the companion paper (holomorphic closure of $E$): the algebra is* dense *in $C(K, \mathbb{C})$, but individual* eml*-trees remain holomorphic. Density is an approximation result, not a representation result.*

## Summary of Proof Status

| Result | Status | Evidence |
|---|---|---|
| Lemma 1.1 (conjugation formula) | Proved | Steps 1–5 above |
| Theorem 3.1 (exact domain $[-\pi, \pi)$) | Proved | Section 3 + SymPy |
| Assumption 4.1 (arithmetic trees) | Open (analytic) | Numerically verified, 60 dps |
| Theorem 5.1 (Stone–Weierstrass) | Conditional on Ass. 4.1 | Complete given assumption |

## References


[1] A. Odrzywołek, *All elementary functions from a single operator*, arXiv:2603.21852v2 (April 2026).

[2] M. H. Stone, *The generalized Weierstrass approximation theorem*, Mathematics Magazine **21** (1948), 167–184.

[3] G. Terzo, *Some consequences of Schanuel's conjecture in exponential rings*, Comm. Algebra **36** (2008), 1171–1189.


 **Repository:** `github.com/antparis/eml_star` **Zenodo:** 10.5281/zenodo.19183008